1.0 Case 1: Introduction - This report explores modifying the James Webb Space Telescope (JWST) bus component for deep space travel. The proposed design will optimize the bus's propulsion system so that it can orbit Pluto based on the existing JWST technology.

2.0 Case 1: Problem statement - The JWST was launched in 2021 and has a propulsion system designed to orbit the second Sun-Earth Lagrange point (L2). However, this system limits the JWST's lifespan by having a finite supply of propellant, which can only sustain a 5-10 year mission. Its ability to capture the universe at the near/mid-infrared spectrum has revolutionized astronomy. As sustainable deep space travel is getting increasingly more feasible, the team is tasked with redesigning the bus component to support sustainable deep space interstellar travel.

3.0 Case 1: Functions, objectives, and constraints - This project aims to achieve one primary function: transform the JWST's current bus into an efficient, sustainable system designed for travel orbiting Pluto, and 4 objectives: 1) Promote long-term sustainability, 2) Maximize power efficiency, 3) Achieve thermal stability, and 4) Remain reliable for a 10+ year mission. Constraints are outlined in Appendix B.

4.0 Case 1: Proposed conceptual design - Currently, the JWST's bus system houses all the electrical, communications, and propulsion systems to keep it operational at L2. The current propulsion system uses 2 pairs of secondary combustion augmented thrusters for Delta V orbit corrections and 8 small mono propellant rocket engines for altitude control and fine adjustments [1]. However, when lengthening its lifespan to sustain a 10+ year mission, the team re-engineered the JWST's propulsion system to create the NovaBus. The NovaBus uses nuclear electrical propulsion (NEP) to power ion thrusters for fine adjustments while orbiting Pluto [Appendix C]. The new design received a score of 9 against the old design's 4, as shown in the weighted decision matrix in Appendix D.

5.0 Case 1: Social, Economic, Environmental Analysis - This design will have a net positive impact on stakeholders[Appendix F]. Less propellant is needed, lowering costs in the newer design [Appendix G][Appendix H]. Material sustainability is analyzed in Appendix E.

6.0 Future Proofing - To construct this design, potential materials used should shield radiation, survive extreme temperatures, self-heal, etc [Appendix I].

7.0 Case 1: Conclusion - Implementing nuclear electric propulsion on the NovaBus allows the JWST to be less limited by its amount of propellant, and therefore more sustainable for deep space travel.

7.0 Case 2: Introduction - As deep space travel becomes increasingly more innovative, communications systems with spacecraft need to be considered when creating cutting-edge new technology. The proposed design will outline a more efficient communications method for interacting with the modified JWST (with the NovaBus bus system) to create a reliable, efficient, and timely solution.

8.0 Case 2: Problem statement - Currently, the Deep Space Network (DSM) relies on 3 global transmission stations, to sustain a reliable signal as the Earth rotates. In the Lucy mission around Jupiter, the DSM network communicated with spacecraft at a rate of 10kbps-50kbps on average [2]. Currently, 1-way radio-signals take 4 hours and 27 minutes to send messages to Pluto [3]. To solve this, the team was tasked with resigning the DSM communications system to contact the modified JWST/NovaBus system from Case 1.

9.0 Case 2: Functions, objectives, and constraints - This project aims to fulfill its primary function: modifying the DSM network to contact the modified JWST/NovaBus telescope in as little time as possible, and 4 objectives: 1) Promote long term sustainability, 2) Large data output, 3) Operate with minimal interference, and 4) Support high wavelength frequencies. Constraints can be found in Appendix L.

10.0 Case 2: Proposed conceptual design - The current DSM system relies heavily on radio waves for interstellar communication. Its structure, with 3 global stations, is incredibly inefficient for deep space travel. For example, the New Horizon's mission, intended to enter a trajectory past Pluto, uses radio frequencies and takes 5+ hours to communicate one-way [4]. The team's proposed design scales the current DSM system to include a Deep Space Optical Communications (DSOC) to increase both transmission speed while using the nuclear energy generated from NovaBus' design as an efficient energy source. This system is fully outlined in Appendix M. This design scores 8.125 against the old design's 6.25 in a weighted decision matrix, shown in Appendix N.

11.0 Case 2: Social, Economic, Environmental Analysis - This design will have a net positive impact on stakeholders[Appendix O]. Cost analysis is in Appendix Q.

12.0 Case 2: Conclusion - Using a DSOC system for deep space communications will increase transmission speed while decreasing the time it takes for a signal to go through.

13.0 References

[1] “JWST Propulsion - JWST User Documentation,” *Stsci.edu*, May 17, 2017. https://jwst-docs.stsci.edu/jwst-observatory-hardware/jwst-spacecraft-bus/jwst-propulsion#gsc.tab=0

[2] M. Cox, C. Cuddy, J. Schneider, T. Kennedy, J. Taylor, and G. Bollendonk, “The Lucy Spacecraft.” Available: https://ntrs.nasa.gov/api/citations/20250001468/downloads/Lucy_Spacecraft_Chapter.pdf

[3] “Exploring Interplanetary Communication Courtesy of NASA / JPL-Caltech / R. Hurt.” Available: https://spacemath.gsfc.nasa.gov/Pluto/PlutoE.pdf

[4] “Final Pluto data transmission received from flyby probe – Spaceflight Now,” *Spaceflightnow.com*, 2025. https://spaceflightnow.com/2016/10/27/final-pluto-data-transmission-received-from-flyby-probe/

[5] A. Price, “Looking into the Past: Space Telescopes and the Law of Outer Space | In Custodia Legis,” *The Library of Congress*, Jan. 24, 2022. https://blogs.loc.gov/law/2022/01/looking-into-the-past-space-telescopes-and-the-law-of-outer-space/

[6] M. D. Rayman and C. E. Garner, “In-Flight Operation of the Dawn Ion Propulsion System Through Completion of the Final Orbit Around Dwarf Planet Ceres,” *Nasa.gov*, Jul. 09, 2018. https://ntrs.nasa.gov/citations/20210008613 (accessed Jan. 24, 2026).

[7] J. Fisher, “NEXT-C Flight Ion System Status,” *Session: Flight Projects, Overviews, and Mission Analysis*, vol. 2020, no. 3604, Aug. 2020, doi: https://doi.org/10.2514/6.2020-3604.

[8] M. Gibson, D. Poston, P. Mcclure, T. Godfroy, M. Briggs, and J. Sanzi, “Kilopower Reactor Using Stirling TechnologY (KRUSTY) Nuclear Ground Test Results and Lessons Learned,” 2018. Available: https://ntrs.nasa.gov/api/citations/20180007389/downloads/20180007389.pdf

[9] C. Webber, “Plasma, Ion-Thrusters, and VASIMR,” 2011. Available: https://www.colorado.edu/faculty/kantha/sites/default/files/attached-files/webber_vasimr.pdf

[10] “Active Thermal Control System (ATCS) Overview,” *BOEING*. Active Thermal Control System (ATCS) Overview

[11] V. S. Jasvanth, A. A. Adoni, V. Jaikumar, and A. Ambirajan, “Design and testing of an ammonia loop heat pipe,” *Applied Thermal Engineering*, vol. 111, pp. 1655–1663, Jul. 2016, doi: https://doi.org/10.1016/j.applthermaleng.2016.07.106.
[12] NASA, “4.0 In-Space Propulsion - NASA,” *NASA*, Mar. 17, 2024. https://www.nasa.gov/smallsat-institute/sst-soa/in-space_propulsion/
[13] G. Woodcock, L. Kos, L. Johnson, J. Jones, A. Trausch, and B. Eberle, “Benefits of Nuclear Electric Propulsion for Outer Planet Exploration.” Available: https://ntrs.nasa.gov/api/citations/20020084995/downloads/20020084995.pdf
[14] Government of Canada, “About the James Webb Space Telescope,” *Government of Canada*, Mar. 30, 2021. https://www.asc-csa.gc.ca/eng/satellites/jwst/about.asp
[15] S. Oleson and T. Packard, “A Deployable 40 kWe Lunar Fission Surface Power Concept.” https://ntrs.nasa.gov/api/citations/20220004670/downloads/40%20kW%20Deployable%20FSP%20Paper_FINAL.pdf?utm_source=chatgpt.com
[16] Canadian Space Agency, “How Webb stays cool,” *Canadian Space Agency*, 2021. https://www.asc-csa.gc.ca/eng/blog/2021/07/23/how-webb-stays-cool.asp
[17] NASA, “Webb FAQs - NASA Science,” *Nasa.gov*, Aug. 12, 2024. https://science.nasa.gov/mission/webb/faqs-full/
[18] “The Role of Nuclear Power and Nuclear Propulsion in the Peaceful Exploration of Space.” Available: https://www-pub.iaea.org/MTCD/publications/PDF/Pub1197_web.pdf
[19] C. G. Doll *et al.*, “Abatement of xenon and iodine emissions from medical isotope production facilities,” *Journal of Environmental Radioactivity*, vol. 130, pp. 33–43, Apr. 2014, doi: https://doi.org/10.1016/j.jenvrad.2013.12.006.
[20] M. Crofton and T. Hain, “The 30 Environmental Considerations for Xenon Electric Propulsion,” 2007. Available: https://electricrocket.org/IEPC/IEPC-2007-257.pdf
[21] strimble, “Importance of Hydrazine Gas Detection in the Aerospace Industry,” *Interscan | Fixed & Portable Industrial Gas Detectors*, Feb. 07, 2024. https://gasdetection.com/articles/importance-of-hydrazine-gas-detection-in-the-aerospace-industry/
[22] W. de Vries, “Impacts of Nitrogen Emissions on Ecosystems and Human health: a Mini Review,” *Current Opinion in Environmental Science & Health*, vol. 21, no. 1, p. 100249, Mar. 2021, doi: https://doi.org/10.1016/j.coesh.2021.100249.

[23] T. V. Koroleva *et al.*, “The environmental impact of space transport,” *Transportation Research Part D: Transport and Environment*, vol. 58, pp. 54–69, Jan. 2018, doi: https://doi.org/10.1016/j.trd.2017.10.013.
[24] E. Sirieys, “Environmental Impact of Space Launches and Societal Response,” 2022. Available: https://dspace.mit.edu/bitstream/handle/1721.1/144846/Sirieys-elwyn-MS-AeroAstro-2022-thesis.pdf?sequence=1&isAllowed=y
[25] M. Thomas, “Award Title: Fusion-Enabled Pluto Orbiter and Lander Report Type: Phase I Final Report Principal Investigator,” 2016. Available: https://www.nasa.gov/wp-content/uploads/2016/04/niac_2016_phasei_thomas_fusiontopluto_tagged.pdf
[26] “Technical Details for the James Webb Space Telescope | Northrop Grumman,” *Northrop Grumman*, 2025. https://www.northropgrumman.com/what-we-do/space/spacecraft/webb-telescope/technical-details
[27] Y. Guo and R. W. Farquhar, “New Horizons Mission Design,” *Space Science Reviews*, vol. 140, no. 1–4, pp. 49–74, Aug. 2007, doi: https://doi.org/10.1007/s11214-007-9242-y.
[28] NASA, “Orbit - NASA Science,” *science.nasa.gov*, Aug. 2024. https://science.nasa.gov/mission/webb/orbit/
[29] NASA, “Pluto: Facts - NASA Science,” *science.nasa.gov*, 2023. https://science.nasa.gov/dwarf-planets/pluto/facts/
[30] A. B.A. , “The Results of Testing and Effectiveness of the Kr-Xe Mixture Application in SPT .” https://electricrocket.org/IEPC/64_3.pdf
[31] “ Website Listing for Aerospace Energy Standard Prices for DoD Customers Effective 1 Oct 2023,” *Defense Logistics Agency*. https://www.dla.mil/Portals/104/Documents/Energy/Standard%20Prices/Aerospace%20Prices/E_2023Oct1AerospaceStandardPrices_230926.pdf?ver=vnDpMqVy1Y_gEIQOsdRo7g%3D%3D
[32] “AEROSPACE ENERGY STANDARD PRICES FOR DOD CUSTOMERS EFFECTIVE 1 OCT 2012.” https://www.dla.mil/Portals/104/Documents/Energy/Standard%20Prices/Aerospace%20Prices/E_2012Oct1AerospaceStandardPrice_160112.pdf

[33] “NASA TECHNICAL M E M O R A N D U M oo t X COMPARISON OF TUNGSTEN AND DEPLETED.” Available: https://ntrs.nasa.gov/api/citations/19690026439/downloads/19690026439.pdf
[34] B. M, “LARGE AND HIGHLY STABLE STRUCTURES MADE OF SiC.” https://www.mersen.com/sites/default/files/files_imported/ecssmet-2012-large-highly-stable-structures-made-of-sic-boostec-mersen.pdf
[35] “The New Space Race for LEO Satellites Supported by Boron Nitride,” Mar. 16, 2022. https://www.bn.saint-gobain.com/blog/new-space-race-leo-satellites-supported-boron-nitride
[36] D. Di-Marco *et al.*, “Dielectric properties of pure alumina from 8 GHz to 73 GHz,” *Journal of the European Ceramic Society*, vol. 36, no. 14, pp. 3355–3361, Nov. 2016, doi: https://doi.org/10.1016/j.jeurceramsoc.2016.05.047.
[37] A. Zaborowska *et al.*, “Absolute radiation tolerance of amorphous alumina coatings at room temperature,” *Ceramics International*, vol. 47, no. 24, pp. 34740–34750, Sep. 2021, doi: https://doi.org/10.1016/j.ceramint.2021.09.013.
[38] A. K. Grain, N. K. Singh, Shubhendra Shivam Maurya, K. K. Pandey, S. K. Ghosh, and Anup Kumar Keshri, “Enhancing high-temperature wear resistance of plasma-sprayed NiCrBSi coatings with nanodiamond reinforcement,” *Diamond and Related Materials*, vol. 150, pp. 111748–111748, Nov. 2024, doi: https://doi.org/10.1016/j.diamond.2024.111748.
[39] B. Sauser, “More Efficient Space Engine Uses Carbon Nanotubes,” *MIT Technology Review*, Dec. 08, 2009. https://www.technologyreview.com/2009/12/08/92672/more-efficient-space-engine-uses-carbon-nanotubes/ (accessed Jan. 24, 2026).
[40] Y. Sun, L. Fu, Z. Fu, A. Shan, and E. J. Lavernia, “Enhanced thermal stability and ductility in a nanostructured Ni-based alloy,” *Scripta Materialia*, vol. 141, pp. 1–5, Jul. 2017, doi: https://doi.org/10.1016/j.scriptamat.2017.07.013.
[41]
V. Anil Kumar, R. K. Gupta, M. J. N. V. Prasad, and S. V. S. Narayana Murty, “Recent advances in processing of titanium alloys and titanium aluminides for space applications: A review,” *Journal of Materials Research*, vol. 36, no. 3, pp. 689–716, Jan. 2021, doi: https://doi.org/10.1557/s43578-021-00104-w.

[42] L. Pernigoni, U. Lafont, and A. M. Grande, “Self-healing materials for space applications: overview of present development and major limitations,” *CEAS Space Journal*, vol. 13, no. 3, pp. 341–352, Apr. 2021, doi: https://doi.org/10.1007/s12567-021-00365-5.
[43] G. Costanza and M. E. Tata, “Shape Memory Alloys for Aerospace, Recent Developments, and New Applications: A Short Review,” *Materials*, vol. 13, no. 8, p. 1856, Apr. 2020, doi: https://doi.org/10.3390/ma13081856.
[44] Q. Zhang, L. Liu, C. Pan, and D. Li, “Review of recent achievements in self-healing conductive materials and their applications,” *Journal of Materials Science*, vol. 53, no. 1, pp. 27–46, Jul. 2017, doi: https://doi.org/10.1007/s10853-017-1388-8.
[45] G. C. Birur, Georg Siebes, T. D. Swanson, and E. I. Powers, “Spacecraft Thermal Control,” *Elsevier eBooks*, pp. 485–505, Jan. 2003, doi: https://doi.org/10.1016/b0-12-227410-5/00900-5.
[46] J. E. Fesmire, “Aerogel insulation systems for space launch applications,” *Cryogenics*, vol. 46, no. 2–3, pp. 111–117, Feb. 2006, doi: https://doi.org/10.1016/j.cryogenics.2005.11.007.
[47] NASA, “9.0 Communications - NASA,” *NASA*, Feb. 12, 2024. https://www.nasa.gov/smallsat-institute/sst-soa/soa-communications/
[48] G. Fountain *et al.*, “The New Horizons Instrument Suite,” *Johns Hopkins APL Technical Digest*, vol. 37, no. 1, 2023, Available: https://www.jhuapl.edu/sites/default/files/2024-09/37-01-Fountain.pdf
[49] M. Cox, C. Cuddy, J. Schneider, T. Kennedy, J. Taylor, and G. Bollendonk, “The Lucy Spacecraft.” Available: https://ntrs.nasa.gov/api/citations/20250001468/downloads/Lucy_Spacecraft_Chapter.pdf
[50] “Final Pluto data transmission received from flyby probe – Spaceflight Now,” *Spaceflightnow.com*, 2025. https://spaceflightnow.com/2016/10/27/final-pluto-data-transmission-received-from-flyby-probe/
[51] G. R. / P. July 17 and 2015, “How exactly does New Horizons send all that data back from Pluto?,” *The Hub*, Jul. 17, 2015. https://hub.jhu.edu/2015/07/17/new-horizons-data-transmission/
[52] “NASA TechPort,” *Nasa.gov*, 2026. https://techport.nasa.gov/projects/33746 (accessed Jan. 24, 2026).
[53] Five Online, “Deep Space Optical Communications - Avantier Inc.,” *Avantier Inc.*, May 22, 2025. https://avantierinc.com/resources/knowledge-center/deep-space-optical-communications/
[54]

W. Thomas Roberts, “Deep-space Optical Terminals (DOT) Ground Laser Transmitter (GLT) Trades and Conceptual Point Design,” Nov. 10, 2010. https://discovery.larc.nasa.gov/pdf_files/31_GLT_Optics.pdf
[55] “New Horizons Phones Home,” *Arrl.org*, 2016. https://www.arrl.org/news/new-horizons-phones-home (accessed Jan. 24, 2026).
[56] “DSOC’s Table Mountain Facility Uplink Laser – Infrared vs. Visible Light,” *NASA Jet Propulsion Laboratory (JPL)*, Sep. 17, 2025. https://www.jpl.nasa.gov/images/pia26662-dsocs-table-mountain-facility-uplink-laser-infrared-vs-visible-light/?utm_source=chatgpt.com (accessed Jan. 24, 2026).
[57] “DSOC’s Table Mountain Facility Uplink Laser – Infrared vs. Visible Light,” *NASA Jet Propulsion Laboratory (JPL)*, Sep. 17, 2025. https://www.jpl.nasa.gov/images/pia26662-dsocs-table-mountain-facility-uplink-laser-infrared-vs-visible-light/ (accessed Jan. 24, 2026).
[58] “DSOC’s Flight Laser Transceiver,” *NASA Jet Propulsion Laboratory (JPL)*, Apr. 06, 2022. https://www.jpl.nasa.gov/images/pia24569-dsocs-flight-laser-transceiver/
[59] “Revitalizing the Deep Space Network to Support NASA’s Growing Space Exploration Program - NASA OIG,” *NASA Office of the Inspector General*, Apr. 09, 2025. https://oig.nasa.gov/news/revitalizing-the-deep-space-network-to-support-nasas-growing-space-exploration-program/
[60] *Jhuapl.edu*, 2016. https://pluto.jhuapl.edu/Mission/Spacecraft/Data-Collection.php
[61] “NASA’s New Horizons Spacecraft Begins Intensive Data Downlink Phase - NASA,” *NASA*, Sep. 04, 2015. https://www.nasa.gov/general/nasas-new-horizons-spacecraft-begins-intensive-data-downlink-phase/
[62] Five Online, “Deep Space Optical Communications - Avantier Inc.,” *Avantier Inc.*, May 22, 2025. https://avantierinc.com/resources/knowledge-center/deep-space-optical-communications/
[63] D. W. H. Johnston, “ Between the Deep Space Network and Deep Space Radio Freguency Interference Protection of Communications,” Jan. 1981. https://ipnpr.jpl.nasa.gov/progress_report/42-62/62R.PDF
[64] “New Horizons Phones Home,” *Arrl.org*, 2016. https://www.arrl.org/news/new-horizons-phones-home (accessed Jan. 24, 2026).

[65] T. Glavich, “Deep Space Optical Communications,” 2015. Accessed: Jan. 24, 2026. [Online]. Available: https://www.nasa.gov/wp-content/uploads/2015/03/tglavich_dsoc.pdf
[66] “Report Concerning Space Data System Standards OVERVIEW OF SPACE COMMUNICATIONS PROTOCOLS INFORMATIONAL REPORT CCSDS 130.0-G-4 GREEN BOOK,” 2023. Available: https://ccsds.org/Pubs/130x0g4e1.pdf
[67] “Deep Space Optical Communications (DSOC),” *NASA Jet Propulsion Laboratory (JPL)*, 2024. https://www.jpl.nasa.gov/press-kits/psyche/dsoc/
[68] “Quick Facts,” *NASA Jet Propulsion Laboratory (JPL)*, 2023. https://www.jpl.nasa.gov/press-kits/psyche/quick-facts/

13.0 Appendices

Appendix A: Figures, Case 1

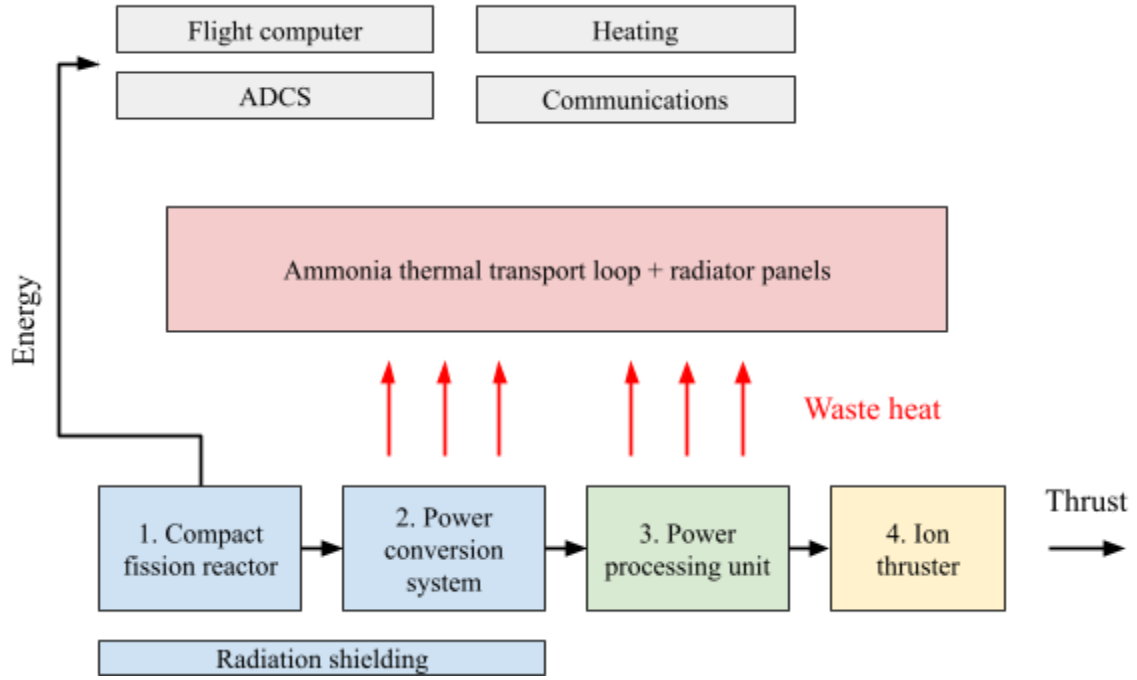


Figure A.1. Process flow diagram for Case 1's proposed design (NovaBus)

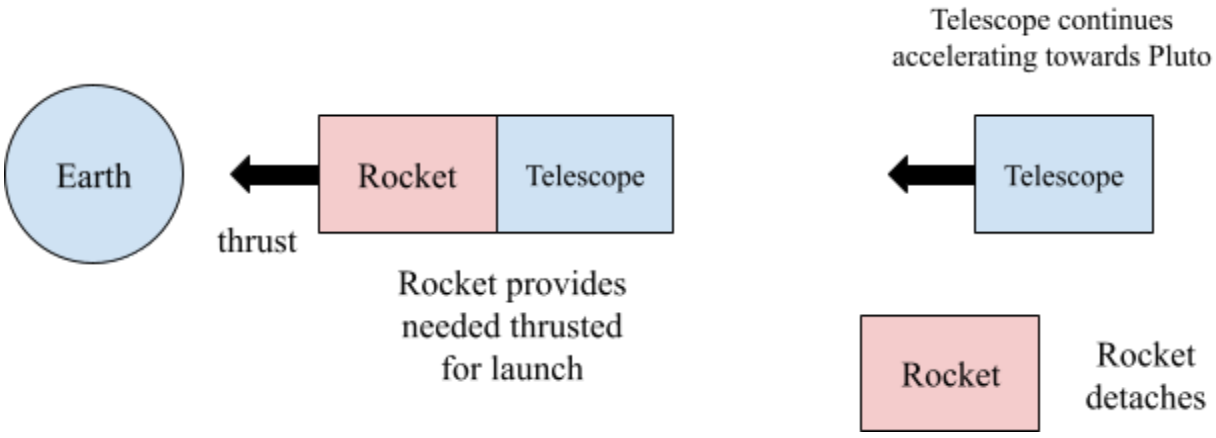


Figure A.2. Representative diagram of thrust provided by the rocket.

Appendix B: Constraints, Case 1

Table B1. NovaBus constraints

Constraint type	Constraints	Justification
Scope	• Delta V orbit corrections and fine adjustments only • Propulsion from Earth to Pluto is out of scope	• The thrust used to get the NovaBus to Pluto's orbit is independent of the Delta V orbit corrections the NEP system will aid
Regulations	• Adhere to the following [5] ○ "Moon agreement" for peacekeeping ○ Convention on International Liability for Damage Caused by Space Objects ○ Convention on Registration of Objects Launched into Outer Space • Ensure launch/orbital safety • Approval from the INSRB • Nuclear power approval	• NovaBus must follow all international space laws and conventions • Since nuclear energy is involved, approval from the DOE must happen before launch
Thrusting capacity	• 11.5 m/s of thrust	• NovaBus' thrusting capacity will be to keep the telescope using orbit correcting Delta V maneuvers • While the exact thrusting capacity is not publicly available (nothing currently orbits Pluto), the thrust capacity was inspired by the ion propulsion system that orbits dwarf planet Ceres [6]
Nuclear energy capacity	• At least 9 kW	• NEXT ion thruster requires 7.4 kW to operate [7] • Upper bounded to account for other bus

		energy requirements
Environmental	See Appendix E for more information	
Cost	See Appendix H for more information	

Appendix C: Technical characteristics details, Case 1

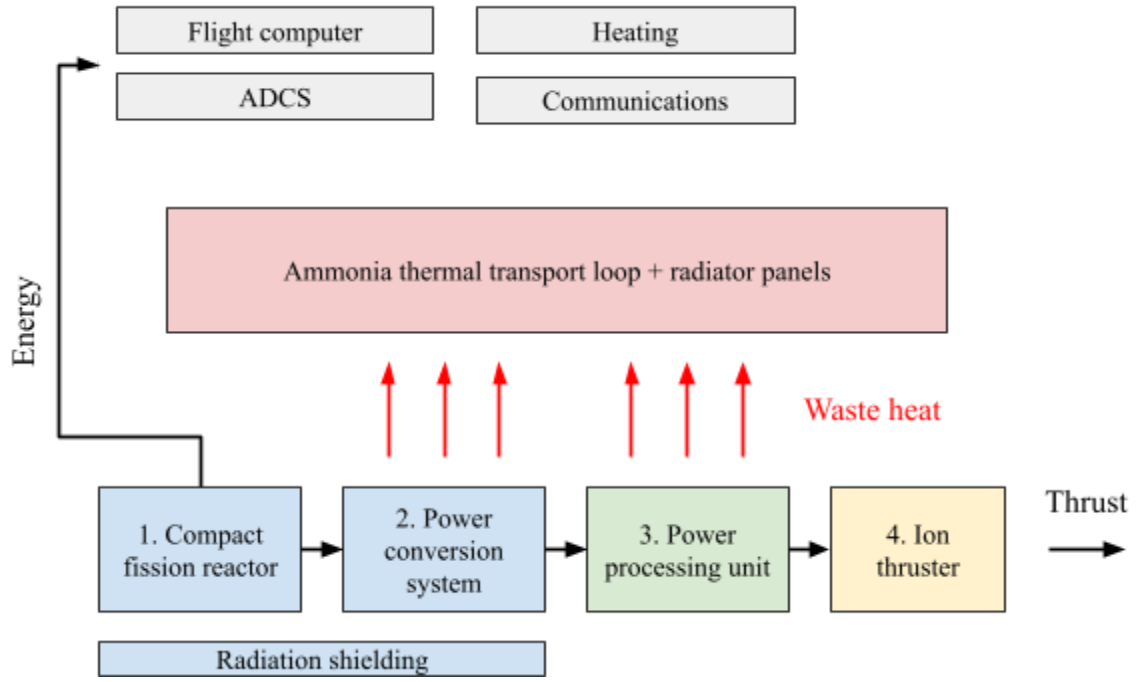


Figure C.1. Process flow diagram for Case 1's proposed design (NovaBus)

Table C.1. Subsystem description of the proposed solution

Subsystem	**Description**	**Justification**
1. Compact fission reactor	The Kilopower Reactor Using Stirling Technology (KRUSTY) fission reactor at <80% capacity [8]	KRUSTY can be scaled to operate from 1-10kWe, will need to operate at 80% capacity max to generate 7kWe for the ion thruster [8]
2. Power conversion system	1.1 Power conversion unit: Converts thermal power into shaft power using sodium heat pipes	High voltage power will be for the thrusters, whereas low voltage power will be for the

	1.2 Power conversion: System converts AC to DC 1.3 Power distribution: Regulated DC power operates the bus load whereas high voltage DC uses the PPU (3) to operate the ion thrusters	flight computer, ADCS, comms, and heating
3. Power processing unit (PPU)	Converts bus power into controlled currents, throttles power	PPU creates high-current pulses of power, which allow the ion thruster to operate [9]
4. Ion thruster	NEXT ion thruster	The NEXT ion thruster only requires <7.4 kW to operate under 4220 ISP at 70% efficiency [7]
Ammonia thermal transport loop	Heat transfer pipes ensure ammonia stays above freezing point. See Figure C.2 for details.	Benchmarked from the ISS, ammonia has a low freezing point, heat pipes ensure ammonia does not reach below that point. See Appendix I for more details [10][11]

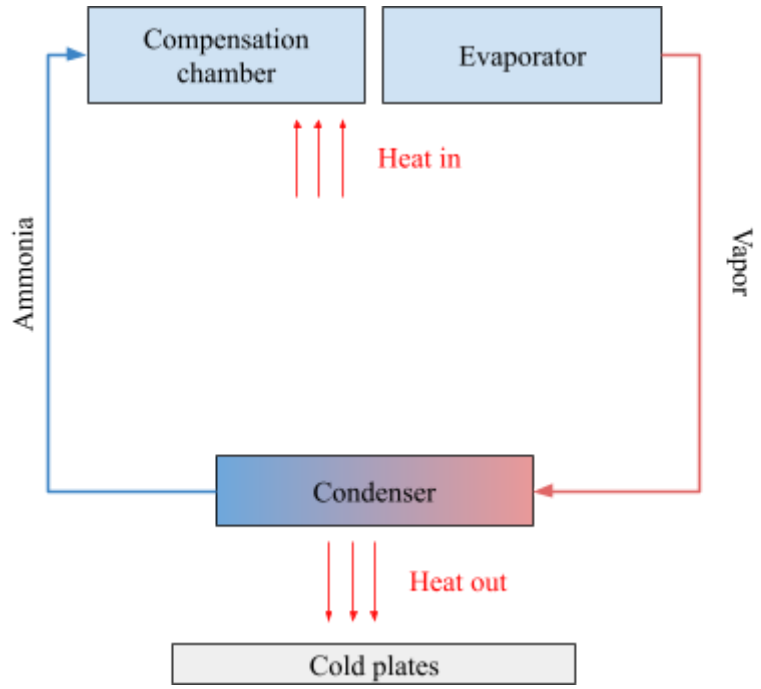


Figure C.2. Process flow diagram for ammonia thermal transport loop

Appendix D: Weighted decision matrix, Case 1

Table D.1. Weighted decision matrix

		Existing design		**Proposed design**	
Objective	**Weight**	**Score (/10)**	**Weighted**	**Score (/10)**	**Weighted**
1) Promote long term sustainability	0.25	3	0.75	9	2.25
Justification		Propellant limits lifespan to only 5-10 years, specific impulse is 200-285 s [source] For material sustainability, refer to Appendix E.		NEP has a ~2000-8000 s specific impulse [12][13] For material sustainability, refer to Appendix E.	
2) Maximize power efficiency	0.25	6	1.5	9	2.25
Justification		Solar panels generate 2,000 W of solar array power [14]		Compact fission reactor that generates 40 kWe [15]	
3) Achieve thermal stability	0.25	7	1.75	9	2.25
Justification		JWST's sunshield protects it from the sun's radiation at L2 [16]		Pluto's climate omits the need for sun radiation protection, ammonia thermal radiator system keep nuclear reactor from causing heat gradients on the proposed design	
4) Remain reliable for a 20+ year mission	0.25	1	0.25	9	2.25
Justification		Will only last at maximum, for 10 years [17]		Nuclear fission reactor, which can be used at 25% to extend lifetime from 10 years to an	

				upper bound of 40 years [18]	
Total	1	-	4.25	-	9

Appendix E: Material Lifetime Sustainability Analysis, Case 1

Material	**Environmental Impact of Production**	**Environmental Impact of Usage**
Xenon	● High energy consumption leads to greenhouse gas emissions [19] ● Requires air separation, and the process produces pollutants [19]	● Inert gas; does not contribute to greenhouse gases and ozone depletion [20]
Hydrazine Fuel	● Negative health effects when exposed to humans and animals [21] ● Accidental discharge into air, soil, and water may occur during any production stage (storage, handling, transport and improper waste disposal) [21] ● Minimal impact because hydrazine degrades quickly in these environments[21]	● Produces toxic emissions of ammonia, hydrogen, and nitrogen gases.[21] ● Produces greenhouse gases that trap heat in an atmosphere. [21]
Dinitrogen Tetroxide	● Discharge into water leads to excess algae growth. ● Pollution of nitrogen oxides are toxic to humans and damaging to plants [22]	● Leakage into soil decreases soil pH [23] ● Causes changes in Ozone ● Forcing agent that changes energy balance of surroundings, can contribute to climate change [24]

Appendix F: Social analysis/implications, Case 1

Table F.1. Stakeholder analysis

Stakeholder	Description/Interest
Government/regulators	● Ensure compliance with international laws and safety standards
Telescope operations team (on Earth)	● Responsible for telescope operations ● Use telescope communications system to operate thrusters
Space agencies	● Funds missions and projects
Material suppliers and manufacturers	● Responsible for testing and manufacturing of different materials
Space research groups	● Will be executing the testing and creation of the NovaBus

Based on the stakeholder analysis table, the following positive and negative impacts on the NovaBus are outlined:

- **Positive impacts**
    - Efficient propulsion system will be created, with less propulsion fluid wasted.
    - Increased research and development about deep space travel.
    - Awareness about nuclear energy sustainability, especially in spacecraft.
- **Negative impacts**
    - High research and development costs before implementation.
    - Ensuring agreement to implement nuclear power in space.

Appendix G: Calculating propellant needed

Tsiolkovsky's rocket equation is used to calculate the propellant needed:

$$\Delta V = I_{sp} g_o ln(\frac{initial\ mass}{final\ mass})$$

The propellant mass can be calculated by subtracting the final mass from the initial mass. It is important to note that ΔV is only relevant for in-orbit course corrections, as the launch is provided by an external rocket.

With the following information:

$\Delta V = 1000 m/s$ [25]

$initial\ mass = 6500 kg$ [26]

$I_{sp} = 4220 s$

$g_o = 9.81 m/s^2$

Substituting into Tsiolkovsky's equation and isolating for final mass results in:

$$m_{prop} = initial\ mass\ (1 - e^{-\Delta v / I_{sp} g_o})$$

$$m_{prop} = (6500)(1 - e^{-1000/(4220*9.81)}) = 155\ kg$$

In comparison, the JWST used 168 kg (370 lbs) of hydrazine fuel and 133 kg (293 lbs) of dinitrogen tetroxide oxidizer, combining to a total of 301 kg [27]. Despite Pluto being significantly further away from Earth than JWST's orbit (5.9 billion kilometres vs 1.5 million kilometres), this proposed design requires less propellant [28][29].

Appendix H: Economic feasibility, Case 1

Exact information on the cost of fission reactors and thrusters is not publicly available. Therefore, the economic feasibility will be calculated using publicly available information on propellant materials.

Table H.1. Cost of implementing NovaBus propellant plan

Propellant	**Unit Cost**	**Unit Cost Adjusted for Inflation**	**Amount Used**	**Total Cost**
Xenon	$1500/kg in 2001 [30]	$2,728.33/kg	155 kg	($2,728.33/kg)* (155 kg) = $422891.15
Hydrazine fuel	$143.53/lb or $65.10/kg in 2019 [31]	$81.98/kg	168 kg	($81.98/kg)*(168 kg) + ($125.23/kg)* (301 kg) = $51466.87
Dinitrogen tetroxide oxidizer	$196.89/lb or $89.31/kg in 2012 [32]	$125.23/kg	301 kg	

In conclusion, the Xenon propellant of NovaBus is more financially affordable than the previous JWST propellant.

Appendix I: Materials used, Case 1

Material Type	Subsystems Used In	Impact on Bus Function
Radiation Shielding		
Depleted uranium, tungsten	Power–Propulsion Module (PPM) Avionics	High density of metal absorbs ionizing radiation (gamma, neutrons) and energetic particles, preventing damage to electronics. [33]
High-Temperature Ceramic Materials		
Silicon carbide (SiC) ceramics	PPM, Ion thrusters, Thermal systems	Maintains strength and thermal conductivity at extreme temperatures, enabling efficient power conversion and heat spreading. [34]
Boron nitride (BN)	Ion thrusters / PPUs	High thermal conductivity helps redirect excess heat from plasma chambers. Ceramic properties are resistant to sputtering (i.e. material wear from ions repeating hitting a surface). [35]
Alumina (Al_2O_3), titania (TiO_2) ceramics	Communications, Electronics	High dielectric strength and radiation resistance protect radio frequency and power electronics. [36][37]
Carbon-Based Nanomaterials		
Plasma-sprayed NiCrBSi coatings with nanodiamond particle reinforcement	Thrusters, Protective coatings	Plasma deposits nanodiamond particles to fill microcracks and prevent material erosion. [38]
Carbon nanotubes (CNTs)	Cathodes, Wiring, Structures	Traditional cathodes require propellant to power electronics and heat them up to obtain sufficient voltage to attract electrons. Instead, carbon nanotubes have a naturally high electrical conductivity. [39]

		High strength-to-weight ratio reduces the mass and energy needed for the satellite. [39]
Lightweight Structural Composites & Alloys		
Nickel-based superalloys	PPM, Thruster components	Retain mechanical strength under thermal cycling and high stress, preventing creep and fatigue. [40]
Titanium alloys	Structural frames, Radiators, Mechanisms	Exhibits high fatigue, fracture resistance, low density and corrosion resistance. [41]
Self-Healing and Adaptive Polymers		
Self-healing polymers Further explored in Appendix J.	Thermal coatings, Antennas, Structures	Embedded healing agents or polymer mobility allow cracks to seal after micrometeoroid or thermal damage. More information in Appendix [42]
Shape-memory alloys	Mechanisms, Deployables	Heat activates components to return to their predefined shapes. [43]
Self-healing conductive inks / CNT wiring	Avionics, Communications	Heat activates material to realign or reconnect across cracks, restoring signal and power paths. [44]
Thermal Control & Insulation Materials		
Ammonia heat pipes / pumped loops	Thermal control system	Phase-change heat transfer moves waste heat to radiators with minimal temperature gradient.[21]
Multilayer insulation (MLI)	Thermal control system	Traps radiative heat transfer, stabilizing sensitive instruments and optics. [45]

Aerogel	Thermal Control System	Low thermal conductivity slows down heat transfer. Can be used as a separator for a warm and cool side. [46]

Appendix J: Self-Healing Materials [42]

Self-Healing Material Type	Key Mechanism
Self-healing ceramics	Embedded particles (ex. SiC, Ti) oxidize at high temperature to fill cracks.
Vitrimers	Reversible bonding ability triggered by heat. Used for adhesives.
Supramolecular Polymers	Reversible physical bonds (ex. H-bonding, ionic) repair damaged areas.
Self-Healing Metals/Alloys	Electropulsing rapidly melts and solidifies metal, creating fine recrystallization zones.

Appendix K: Figures, Case 2

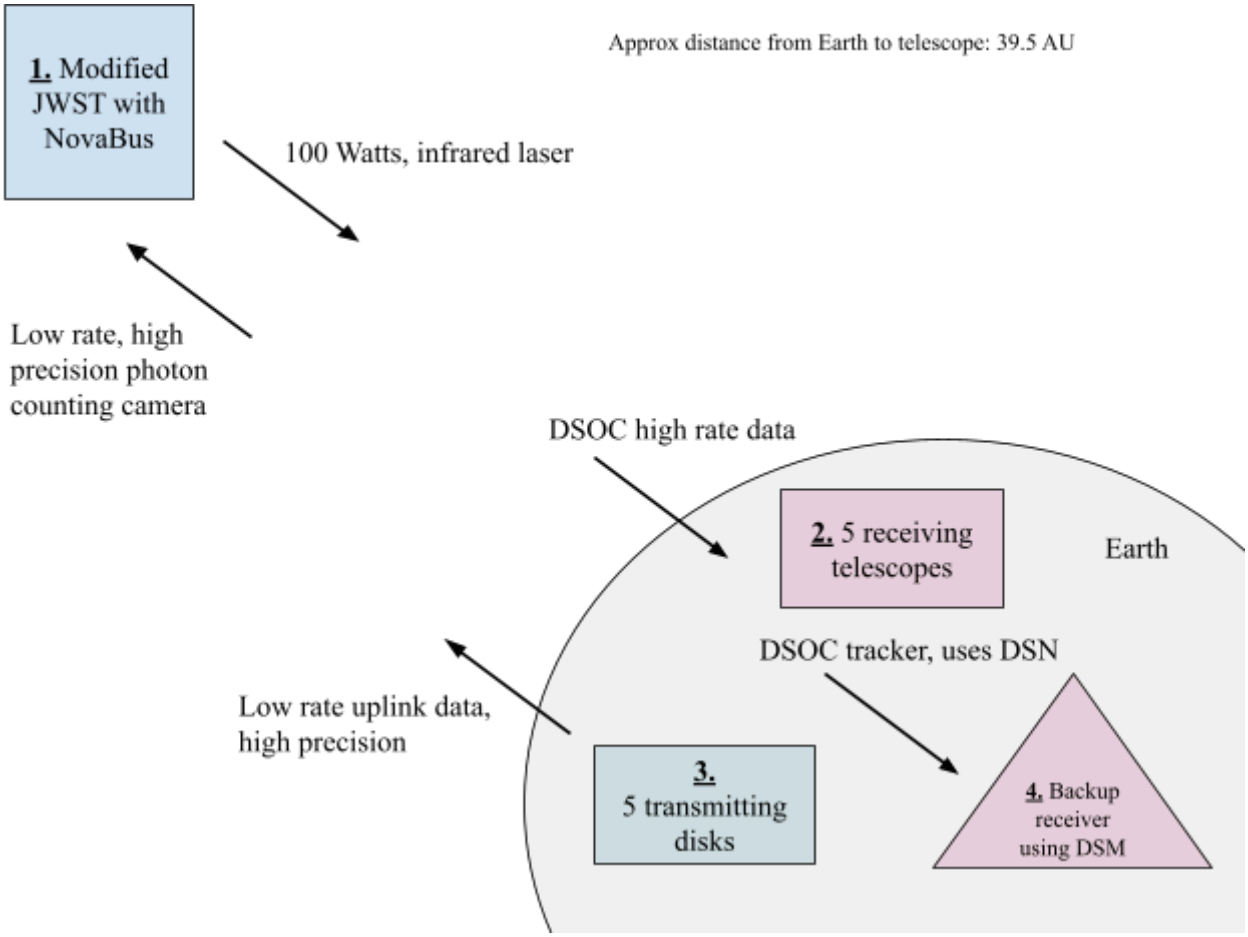


Figure K.1. Process flow diagram for Case 1's proposed design

Appendix L: Constraints, Case 2

Table L.1. Proposed design constraints

Constraint type	**Constraints**	**Justification**
Scope	- Add on to existing DSM architecture - Implement as many existing design principles as possible	- DSM architecture, although slow, can serve as a backup for new designs
Regulations	- Follow international treaties (ISED) [47] - Not interfere with other space communications systems	- Proposed design must follow all international space laws and conventions
Interference	- <5 signal-to-noise ratio (SNR)	- Benchmarked from the New Horizon's SNR [48]
Transmission speed	- <5 hours one direction - Faster than 10-50 kbps [49]	- New Horizons' speed was used as a benchmark (5 hours, 8 mins for a 3.4 billion mile distance) [50] - Lucy mission reached an average of 10-50kbps

Appendix M: Proposed solution, Case 2

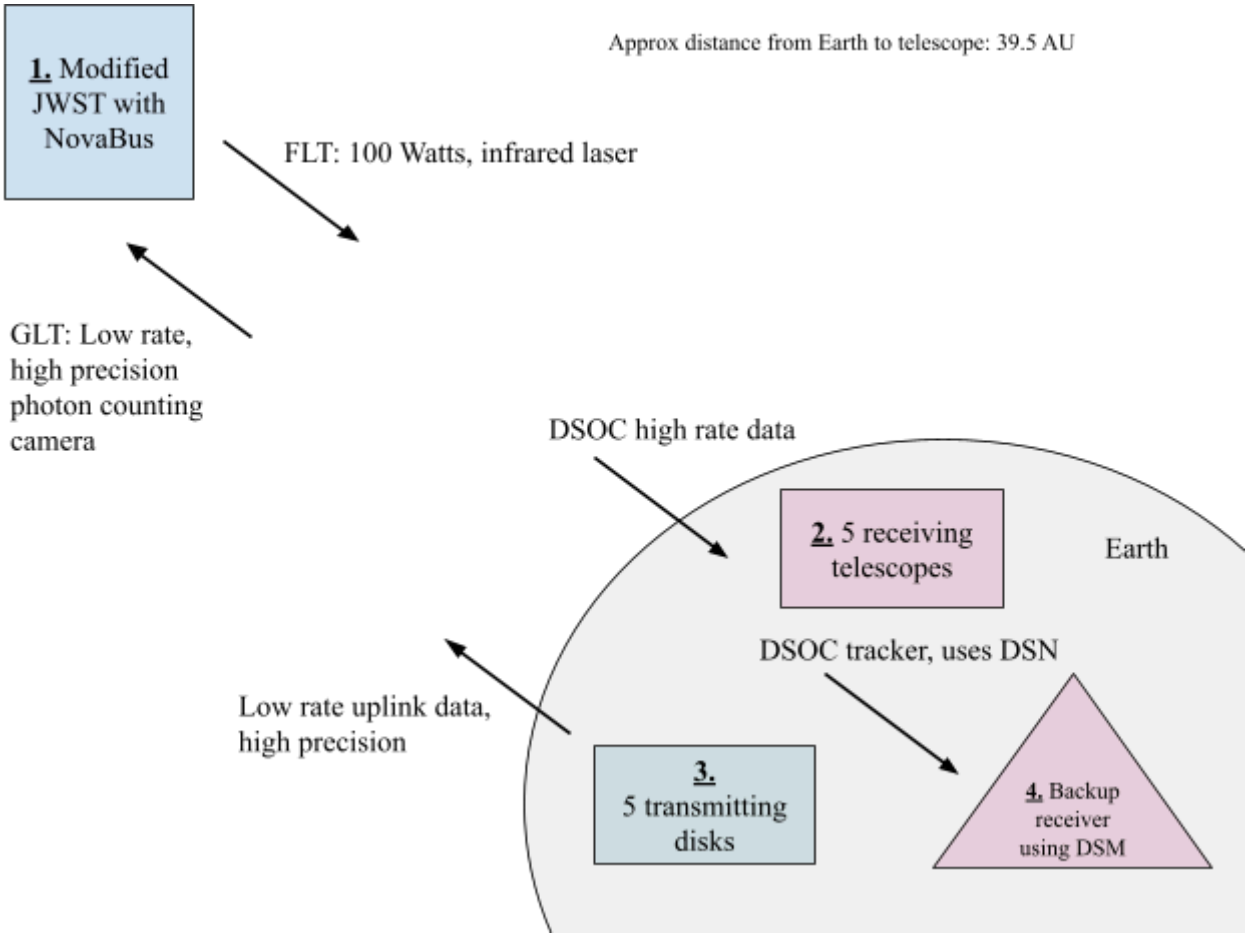


Figure M.1. Proposed design for Case 2. Numbers do not indicate order/hierarchy and are purely used for discrimination in Table M.1.

Table M.1. Proposed design specifications for Case 2

Subsystem	**Description**	**Justification**
1. Modified JWST with NovaBus	1.1 Flight laser transceiver system (FLT): 1550 nm infrared laser with 100 watt transmission capacity, short pulse [51] [52][53] 1.2 Ground laser transmitter (GLT): Accurately pointed beam for photon counting [54]	Compared to the New Horizon's 12W transmission system, which sends 1 kbps to Earth. This is highly limited due to its generator, which has decayed to allowing only 12W of transmission vs the original 24W [55].

	1.3 Vibration prevention system: FLT is mounted on actuators and struts to stabilize vibrations	The NovaBus' nuclear power generation has more power capacity than that of the New Horizon's craft, causing 100W to be used for the optics system.
2. 5 receiving telescopes	High-rate DSOC receiving telescopes in 5 locations within 300 mile radius.	Existing DSM scalability allows an increased amount of telescopes to correlate to higher reception rate [56]
3. 5 transmitting disks	High precision infrared receivers, increased amount increases capacity	Benchmarked with the receiver from the Psyche mission [57]
4. Backup receiver	Use existing DSM architecture to track DSOC downlink	Already implemented for the Psyche mission, good safety net for ensuring minimal data loss [58]

Appendix N: Weighted decision matrix, Case 2

Table N.1. Weighted decision matrix

		Existing design		**Proposed design**	
Objective	**Weight**	**Score (/10)**	**Weighted**	**Score (/10)**	**Weighted**
1) Promote long term sustainability	0.25	7	2.75	8.5	2.125
Justification		NASA has plans to upgrade the DSN infrastructure by replacing its six antennas with new ones, thus extending its life. However, this project is currently 5 years behind schedule [59].		The NovaBus' nuclear power generator will sustainably supply power to the communications systems. A failure in power generation is what caused the New Horizon's system to only work at half its intended efficiency [60].	
2) Large data output	0.25	3	0.75	8	2

Justification		New Horizon, a satellite near Pluto, has a download rate of 1-4 kbps [61].		Having 100W of power to operate the FLT significantly increases the output capabilities, which is where current deep space technology is limited [62]	
3) Operate with minimal interference	0.25	4	2	7	1.75
		Radio frequency is incredibly prone to interference (clouds, radiation) causing it to be inefficient at deep space communications [63]		Infrared-based communication systems are still susceptible to interference and must work around it	
4) Support high wavelength frequencies	0.25	3	0.75	9	2.25
		Deep space technology, such as the New Horizons telescope transmits 1 kbps on average [64]		Having a nuclear power source power a 100W generator is an incredible advantage that allows DSOC infrared lasers to have a wavelength of 1.55 microns [65]	
Total	1	-	6.25	-	8.125

Appendix O: Stakeholders Analysis, Case 2

Stakeholder	**Description/Interest**
Government/regulators	• Ensure compliance with international laws and safety standards
Telescope operations team (on Earth)	• Responsible for telescope operations
Space agencies	• Funds missions and projects
Material suppliers and manufacturers	• Responsible for testing and manufacturing of different materials
Deep Space Network	• Engineers maintain the system 24/7, sending messages as

operators	needed.

Based on the stakeholder analysis table, the following positive and negative impacts of the design are outlined:

- **Positive impacts**
    - Strong, collective global interest in the well-being of the project.
    - Increased research and development on deep space communication.
    - Developments lead to improvements in communications systems overall.
- **Negative impacts**
    - Many global protocols that the design needs to adhere to, as seen in Appendix P.
    - Numerous stakeholders can lead to potential conflicts of interest in development and funding.

Appendix P: Overview of Space Communication Protocols [66]

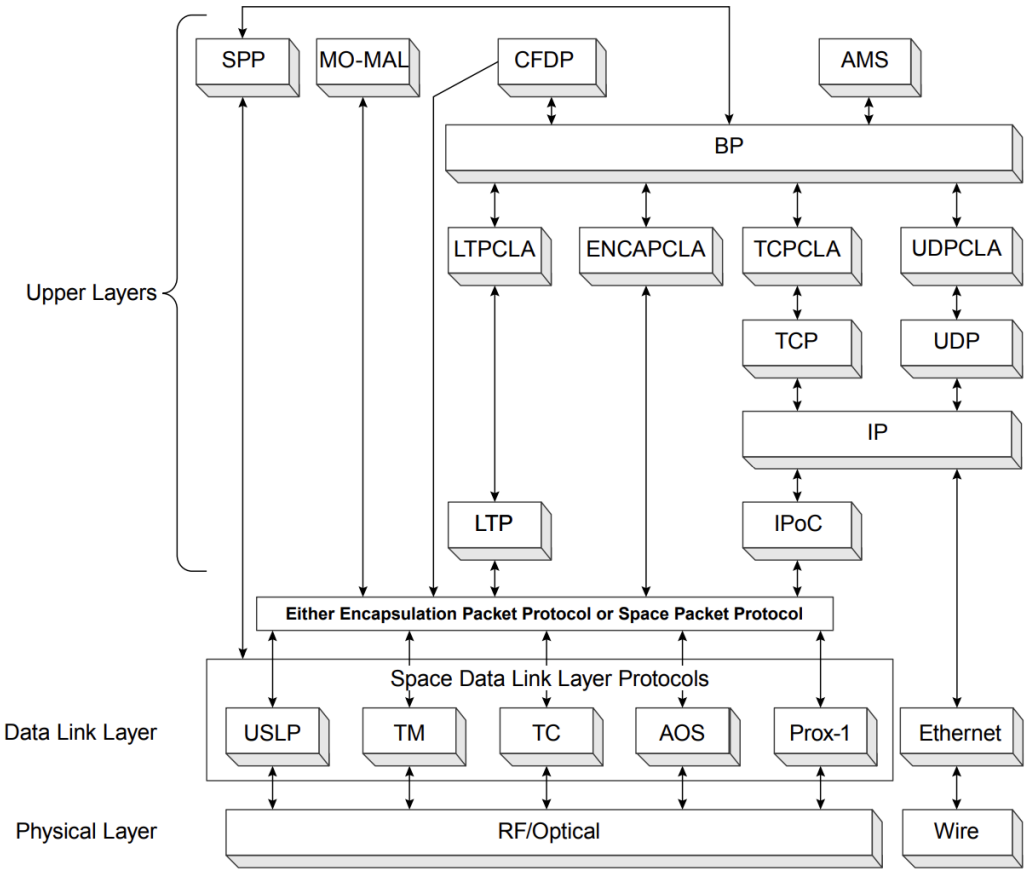


Figure P.1. Model of relationships and hierarchy between space communication protocols. [same link as above]

Layer	Protocols	
Physical Layer	**Radio Frequency and Modulation Systems** ● Omnibus for space links between ground stations and spacecraft. ● Ex. Between spacecraft and ground stations. **Proximity-1 Space Link Protocol** ● Recommendations for proximity space links. ● Ex. Landers, rovers, orbiters, relays.	
Data Link Layer	**Data Link Protocol Sublayer:** Techniques for employing data units called	● **Telemetry (TM) Space Data Link Protocol** ○ Usually for sending telemetry to a ground station from a spacecraft ○ On a return link

	Transfer Frames to transmit data units supplied by the higher layer via a point-to-point space link.	• **Telecommand (TC) Space Data Link Protocol** ○ Usually for sending commands to a spacecraft from a ground station ○ On a forward link • **Advanced Orbiting Systems (AOS) Space Data Link Protocol** ○ Can be used on a return link only, or on both forward and return links ○ Supports two-way, faster communications, like audio and video, between a spaceship and the ground are required. • **Proximity-1 Space Link Protocol—Data Link Layer** ○ For proximity space links ■ Bi-directional, short-range, mobile or fixed radio links. ○ Usually used to communicate between orbiting constellations, orbiting relays, landers, rovers, and fixed probes. • **Unified Space Link Protocol (USLP)** ○ Can be used utilized by space missions across communications lines for space-to-ground, ground-to-space, or space-to-space.
	Synchronization and Channel Coding Sublayer:	• **TM Synchronization and Channel Coding** ○ Used with the TM or AOS Space Data Link Protocol or USLP. ○ Adds error correction and frame synchronization to TM/AOS/USLP frames for noisy downlinks. • **TC Synchronization and Channel Coding** ○ Used with the TC Space Data Link Protocol or USLP. ○ Provides BCH error-correcting coding and synchronization for TC/USLP command uplinks, with optional validation. • **Proximity-1 Space Link Protocol—Coding and Synchronization Layer**

		○ Used with the Proximity-1 Space Link Protocol or USL. ○ Implements convolutional/LDPC coding and CRC for short-range bidirectional links. ● **Flexible Advanced Coding and Modulation Scheme for High Rate Telemetry Applications (SCCC)** ○ For missions requiring very high data rates ○ Ex. Earth observation. ● **CCSDS Space Link Protocols over ETSI DVB-S2 Standard (DVB-S2)** ○ Reduces cost and development time for high-speed downlinks.
Network Layer	● **CCSDS-recognized Internet Protocol datagrams** ● **The Delay Tolerant Network architecture's Bundle Protocol** ○ Implements DTN store-and-forward routing for intermittent, long-delay space links. ○ Core protocol for deep-space, lunar, and disrupted environments.	
Transport Layer	● **Space Communications Protocol Standards (SCPS) Transport Protocol** ○ Extensions to TCP/UDP for space environments. ○ End-to-end reliable transport. ● **Transmission Control Protocol (TCP)/ User Datagram Protocol (UDP)** ○ Traditional transport protocols ○ Used over IP over CCSDS links ● **Licklider Transmission Protocol** ○ Provides optional reliability for DTN links ○ Underlying transport for Bundle Protocol	
Application Layer	● **Asynchronous Messaging Service (AMS)** ● **CFDP** ● **Lossless Data Compression** ● **Image Data Compression** ● **Multispectral & Hyperspectral Image Compression**	

	• **Space Packet Protocol** • **Message Abstraction Layer (MAL) Space Packet Transport Binding and Binary** • **Encoding**

Appendix Q: Cost Analysis of Design, Case 2

Table Q. Ratio of transmission power between Psyche and OCTL.

Transmission	**Ratio of transmission power**
Transmission from Psyche: 4 watt [67]	0.0008
Transmission from OCTL: 5 kWatt [67]	0.9992

The cost of developing and operating DSOC is about $206 million [68]. This leads to a cost breakdown of $164,800 for Psyche transmission, and around $205.8 million for OCTL transmission. In this new proposed design, Psyche transmission is increased to 100 watts. Additionally, both transmission and receiving devices are increased from 1 to 5. The new cost breakdown is shown as:

Transmission from Psyche: $164,800 * 100/4 * 5 = $206 million

Transmission from OCTL: $205,835,200 * 5 = $1029.2 million

The cost for the new proposed design is estimated to be $1.235 billion. This is more expensive than the previous design, but it will also provide greater data transmission power. Therefore, this is a trade-off with a positive benefit.